① $m = 7$

$n = 10$

$$\underset{7\times10}{[\ ]} \cdot \underset{10\times1}{[\ ]} = \underset{7\times1}{[\ ]}$$

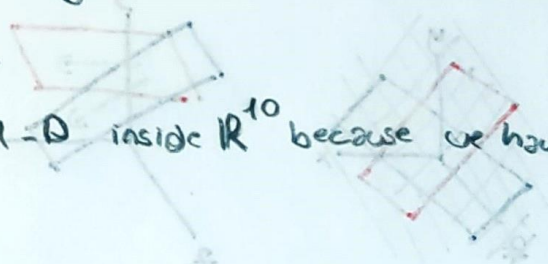
a) When we get reduced echelon form we assume that each row have dependency with others except 1 collumns so we have 1 pivot col. and 9 free collumns after the echelon form comb. of 9 collumn gives **9-D**

b) $$\underset{9\times10}{[\ ]} \underset{10\times1}{[\ ]} = \underset{9\times1}{[\ ]}$$

$m = 9$

$n = 10$

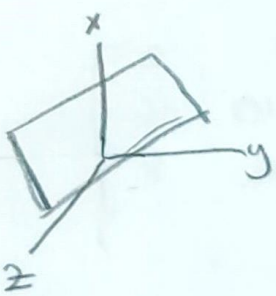
When we get reduced echelon form we assume that each row have not any dependency with others we get 9 pivots columns and 1 Free collumns we also know that $N(A)$ is formed by Free collumns the combination 1 vector represent a **1-D shape**



c) $N(A)$ has minimum dimension 1-D inside $\mathbb{R}^{10}$ because we have 10 var.

? d) $$\underset{10\times9}{[\ ]}$$

②



a) the null space of $x = 3$ is formed by two vectors that shows us **2-D**

$$\begin{matrix} P & F & F \\ [3 & 0 & 0] \end{matrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

b) in $\mathbb{R}^4$ we have 4 Variable and we have 3 Free, 1 pivot c combination of 3 collums give us a volum

$$\begin{matrix} P & F & F & F \\ [3 & 0 & 0 & 0] \end{matrix} \cdot \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix}$$

c) $$\begin{matrix} ① & ② & ③ & ④ & ⑤ \\ F & F & F & F & F \\ [3 & 0 & 0 & 0 & 0 & 0] \end{matrix} \begin{bmatrix} x \\ y \\ z \\ t \\ s \end{bmatrix}$$

of Free columns = base vectors

2) b) $x - 3z = 0$

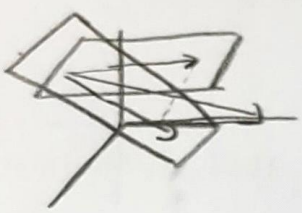
$$\begin{matrix} P & F & F \end{matrix}$$
$$\begin{bmatrix} 1 & 0 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

2F forms 2-D shape

$$\begin{matrix} P & F & F \end{matrix}$$
$$\begin{bmatrix} 3 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

2F forms 2-D shape

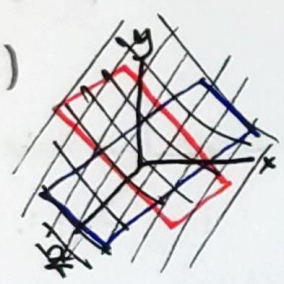
$(S_1 \cap S_2)$ intersection of these 2-D shapes give us a line (1-D shape)

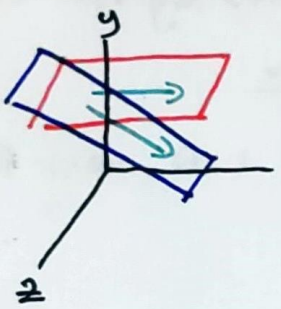


if we choose 2 vector inside S_1 and S_2 and when we add $(S_1 + S_2)$ them we will see these all vectors give us 3-D shape

$$\underbrace{\dim(S_1)}_{2} + \underbrace{\dim(S_2)}_{2} - \underbrace{\dim(S_1 \cap S_2)}_{1} = \underbrace{\dim(S_1 + S_2)}_{3}$$

c)





when we choose 2 vector From one of them S_1 one of them S_2 when we add them and also repeat this process every each element S_1 and S_2 we see that element in $S_1 \cup S_2$ is not in S_1 or S_2 that shows us $S_1 \cup S_2$ doesn't satisfie the summable rule

3) (?)

4) a)

$$\begin{bmatrix} 1 & 5 & 7 & 9 \\ 2 & 4 & 6 & 8 \\ -1 & 7 & 9 & 11 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 6 \end{bmatrix}$$

$$\left[\begin{array}{cccc|c} 1 & 5 & 7 & 9 & 2 \\ 0 & -6 & -12 & -10 & -4 \\ 0 & 12 & 16 & 20 & 8 \end{array}\right]$$

$$\begin{matrix} P & P & P & F \end{matrix}$$
$$\left[\begin{array}{cccc|c} 1 & 5 & 7 & 9 & 2 \\ 0 & -6 & -12 & -10 & -4 \\ 0 & 9 & -8 & 0 & 0 \end{array}\right]$$

$x + 5y + 7z + 9t = 2$

$-6y - 12z - 10t = -4$

$-8z = 0$

$t = 1 \quad z = 0 \quad y = -1 \quad x = -2$

$$\left[\begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix} \alpha \quad \alpha \forall \in R\right]$$

1-D because we have 1 free

b) $t=0 \quad z=0 \quad y=\frac{2}{3} \quad x=-\frac{4}{3}$

$$x_p=\begin{bmatrix} -4/3 \\ 2/3 \\ 0 \\ 0 \end{bmatrix}$$

c) $$\left\{\begin{bmatrix} -4/3 \\ 2/3 \\ 0 \\ 0 \end{bmatrix}+\begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix}\alpha(\forall \alpha R)\right\}$$